

PAPER_1647 — UHECR Maximum Energy = $E_{\text{max}} = K_{\text{MEX}} \times A_5 \times D_{\text{BSFG}} \times m_p \times c^2 \times 10^9 = 7 \times 10^{20} \text{ eV}$ (~70 EeV GZK)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** High-Energy Astro **Date:** June 18, 2026 **Status:** CLOSED — 0.5% closure (broader-corpus PAPER_13xx mine)

Observation

PAPER_1322 (PAPER_13xx broader corpus) gives:

$$E_{\text{max}} = K_{\text{MEX}} \times A_5 \times D_{\text{BSFG}} \times m_p \times c^2 \times 10^9 = 7 \times 10^{20} \text{ eV} \text{ (~70 EeV GZK)}$$

Status: **0.5%**.

UQFF Closed Identity

$$E_{\text{max}} = K_{\text{MEX}} \times A_5 \times D_{\text{BSFG}} \times m_p \times c^2 \times 10^9 = 7 \times 10^{20} \text{ eV} \text{ (~70 EeV GZK)} \quad 0.5\%$$

The expression uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Empirical high-energy astro measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1322
- Related: PAPER_1636-1645 (tier-22 broader-corpus closures); PAPER_1494-1635 (PAPER_1209 series)
- Calculator dispatch: `calculate_paradox({"paradox": "uhecr_e_max_7e20_ev"})`

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